Unit 3 Linux Documentation

Course code LX13
Linux Basics

Objectives

After completing this unit, students should be able to:

- Use the `man` command to view information about Linux commands
- Describe the use of `info`
- Describe the HOWTO and FAQ documentation
- Explain the importance of Internet for gathering information about Linux

The man Command

- With the `man` command you can read the manual page of commands.
- Manual pages are stored in /usr/man
- The manual page consists of:

Name	The name of the command and a one-line description
Synopsis	The syntax of the command
Description	Explanation of how the command works and what it does
Files	The files used by the command
Bugs	Known bugs and errors
See also	Other commands related to this one

man Example

```
$ man finger
Formatting page, please wait...
NAME
  finger - user information lookup program

SYNOPSIS
  finger [-lmsp] [user ...] [user@host]

DESCRIPTION
  The finger command displays information
  about the system users.

Options are:
  -s Finger displays the user's login name,
  :
```

man Example (2)

- The **-k** option of the `man` command allows you to print out one-line descriptions of all entries which match the given keyword.
- The **apropos** command is equivalent to **man -k**

```
$ man -k print
arch (1)         - print machine architecture
date (1)         - print or set the system date and time
logname (1)      - print user's login name
lpc (8)          - line printer control program
lpd (8)          - line printer spooler daemon
lpr (1)          - off line print
lprm (1)         - remove jobs from the line printer spooling queue
```

man Sections

- Manual pages are divided in 8 sections:
  1. User commands
  2. System calls
  3. Libc calls
  4. Devices
  5. File formats and protocols
  6. Games
  7. Conventions, macro packages and so forth
  8. System administration
- Certain subjects appear in multiple sections
- To select correct section, add section number:
  man 1 passwd (about the passwd command)
  man 5 passwd (about the passwd file)

The info Command

- A program for reading documentation.
- Sometimes a replacement for manual pages
- Information for info is stored in **/usr/info**
- Some info commands

`space`	next screen of text
`delete`	previous screen of text
n	next node
p	previous node
q	quit info

info Example

```
# info pwd
```

```
File: sh-utils.info,  Node: pwd invocation,  Next: stty invocation,  Up: Workin\
g context

`pwd': Print working directory
============================

   `pwd' prints the fully resolved name of the current directory.  That
is, all components of the printed name will be actual directory
names--none will be symbolic links.

   Because most shells have a built-in command by the same name, using
the unadorned command name in a script or interactively may get you
different functionality than that described here.

   The only options are a lone `--help' or `--version'.  *Note Common
options::.

--zz-Info: (sh-utils.info.gz)pwd invocation, 17 lines --All-----------------------
Welcome to Info version 2.18. "C-h" for help, "m" for menu item.
```

The --help Option

- Another way of getting help about a command
- Help is built in the command itself
- List of all options provided by command

```
$ who --help
Usage: who [OPTION]... [ FILE | ARG1 ARG2 ]

 -h, --heading  print line of column headings
 -m             only hostname and user
                associated with stdin
 -q, --count    all login names and number of
                users logged in
     --help     display this help and exit
     --version  output version information and exit
```

FAQ

- Give answers to **F**requently **A**sked **Q**uestions.
- Are ASCII files stored in **/usr/doc/FAQ**.
- Do not describe specific actions but give answers to questions like:
  - What is Linux?
  - How much memory does Linux need?
  - Where can I get the HOWTOs and other documentation?
  - Can Linux access SMB file systems?
  - Can I run MS Windows programs under Linux?
  - How can I undelete files?
  - How do I make a bootable floppy?
  - and many more questions.

HOWTO Documents

- Documents which describe in detail a certain aspect of configuring or using Linux.
- Detailed information about how to perform a given task
  - Install PCMCIA support
  - Kernel compilation
  - Dual boot with other operating systems
- HOWTO documents are text files in **/usr/doc/HOWTO**
  - Need to be installed manually
- On the Internet:
  - **http://www.linuxdoc.org/index.html**

HOWTO Example

```
$ less /usr/doc/HOWTO/XFree86-HOWTO
The Linux XFree86 HOWTO
by Eric S. Raymond
v5.8, 16 August 1998

This document describes how to obtain,
install, and configure version 3.3 of the
XFree86 version of the X Windows System
(X11R6) for Linux systems. It is a
step-by-step guide to configuring XFree86
on your system.

Table of Contents
:
```

HTML Documentation

- Certain programs also offer HTML documentation
- Usually stored in /usr/doc/<programname>
- Can be read with any browser
  - netscape
  - lynx

Internet

- All Linux documentation is available on the Internet.
- For information look at:
  - http://www.linux.org
  - http://www.redhat.com
  - http://www.caldera.com
  - http://www.pht.com
  - http://www.xfree86.org
  - http://www.kernel.org
  - http://lwn.net
  - http://www.ugu.com
  - and many more
- Usenet news:
  - comp.os.linux.*

Checkpoint

1. Name some of the places where you can get information about Linux and its commands on your system.

a. manual pages
b. info command
c. HOWTO documents
d. FAQ
e. /usr/doc
f. All above

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2. What are HOWTOs?

Unit Summary

- The `man` command can be used from the command line to view the proper syntax of Linux commands.
- For manual pages that are out-of-date, you can use the `info` command to view the syntax.
- Specific system administration tasks are described in the HOWTO documents.
- Commonly asked questions about Linux are answered in the FAQ.
- The Internet is the place for the latest information about Linux